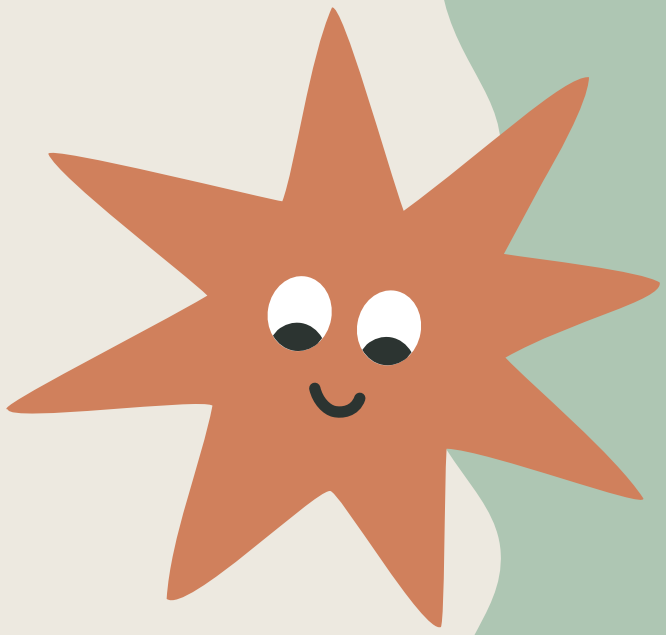
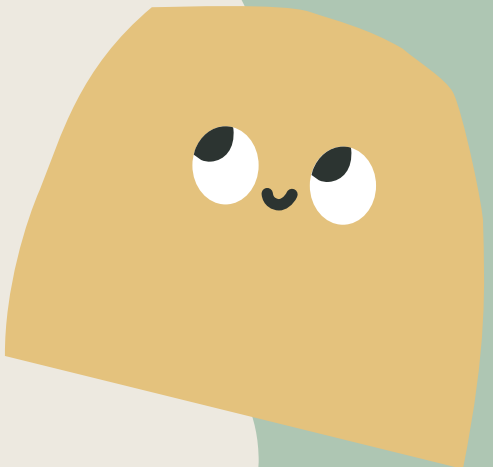
Intelligent Agents

Lecture 3





Learning Outcomes (Lecture 1)

➢At the end of this lecture, students will be able to:

- Define an **agent** in Artificial Intelligence.
- Explain the difference between an ordinary agent and an intelligent agent.
- Explain the relationship between an agent and its environment.
- Understand **sensors** and **actuators**.
- Explain perception and action in an intelligent system.
- Understand the concept of an **agent function**.
- Explain what a **rational agent** is.
- Understand the concept of **rationality**.
- Explain the role of a **performance measure**.
- Describe a simple intelligent agent using a complete example.

Prepared By: Ehtisham Rasheed

What is an Agent?

- In everyday language, an **agent** is someone or something that performs an action on behalf of someone else
- For example:
  - A travel agent helps a customer arrange travel
  - A real-estate agent helps a client buy or sell property
  - A software agent performs tasks automatically
- In Artificial Intelligence, the definition is slightly different
  - An agent is anything that can perceive its environment through sensors and act upon that environment through actuators

Prepared By: Ehtisham Rasheed

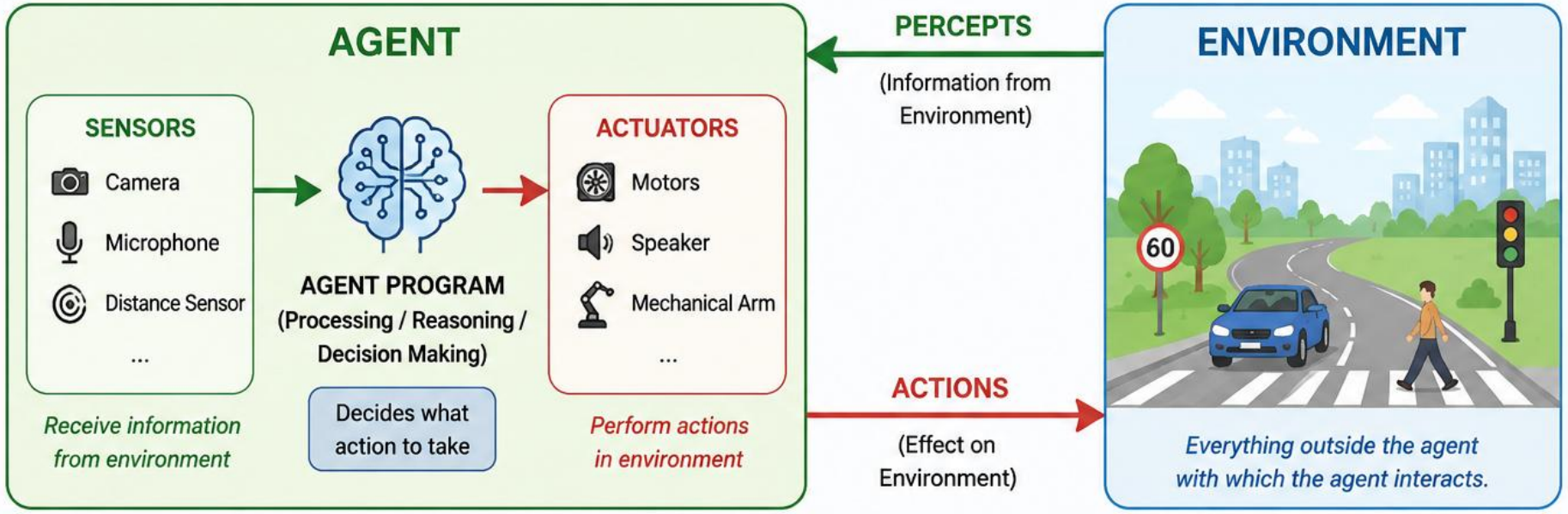
What is an Environment?

- An environment is the **world or situation in which an agent operates**
- For example:
  - Self driving car
  - Chess AI
  - Vacuum-cleaning robot
- The agent receives information from the environment and performs actions in that environment

Prepared By: Ehtisham Rasheed

Agent-Environment Interaction

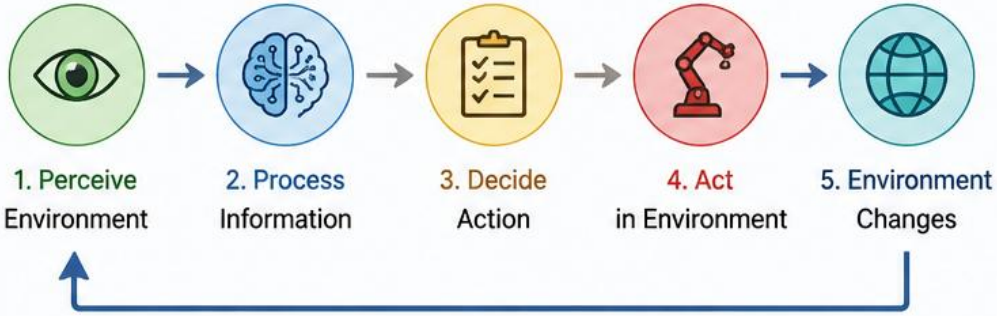
An agent **perceives** its environment through sensors and **acts** upon that environment through actuators.



KEY POINTS

- The agent **SENSES** the environment.
- The agent **THINKS / REASONS** using the percepts and its knowledge.
- The agent **ACTS** to change or affect the environment.

THE INTERACTION CYCLE



EXAMPLES

- Self-driving car
- Vacuum cleaning robot
- Chess AI
- Chatbot
- Recommendation system
- Email spam filter

What is Perception?

- **Perception** means obtaining information about the environment
- For humans, perception happens through our senses:
  - Eyes, ears, nose, skin, tongue
- For an AI system or robot, perception can come from:
  - Cameras, microphones, temperature sensors, distance sensors, GPS, radar,
- For example, a self-driving car's camera may detect:
  - There is a pedestrian ahead
- This information becomes part of the agent's **perception**

Prepared By: Ehtisham Rasheed

What are Sensors?

- A **sensor** is a device or mechanism that allows an agent to obtain information from its environment
- Sensors are the means through which an agent perceives its environment

Agent	Sensors
Human	Eyes, ears, skin
Robot	Camera, microphone, distance sensor
Self-driving car	Cameras, radar, LiDAR, GPS
Smart thermostat	Temperature sensor
Smartphone	Camera, microphone, GPS, accelerometer
Chess AI	Game-board state/input

Rasheed

What are Actuators?

- An actuator is a mechanism through which an agent performs an action on its environment
- Robot actuators:
  - Motors, robotic arm, wheels
- Self-driving cars actuators:
  - Steering, brakes, accelerator
- Human actuators:
  - Muscles
- Software agent actuators:
  - A software agent might perform actions such as:
  - Send email, display message, place an order, update a database etc

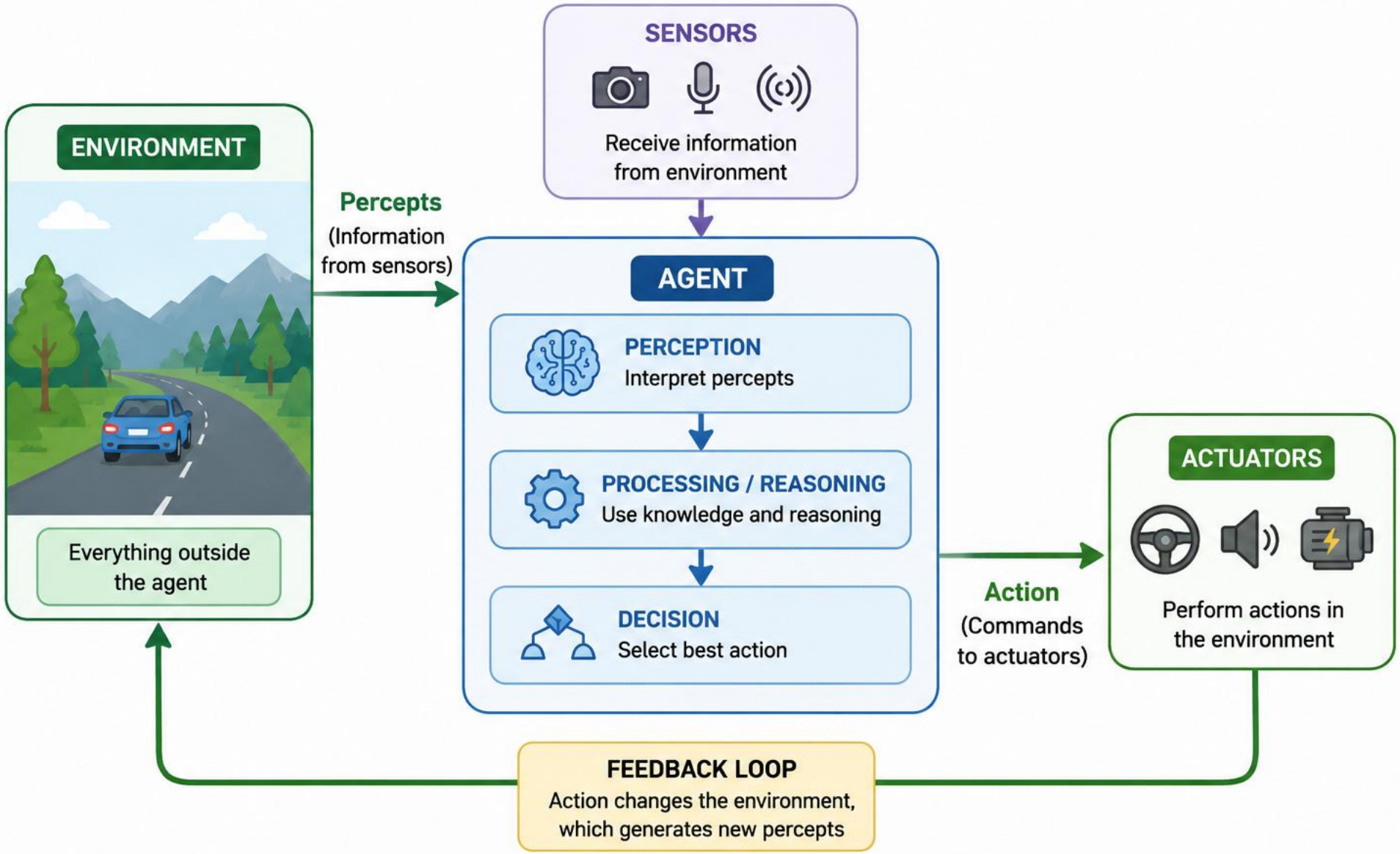
Prepared By: Ehtisham Rasheed

What is an Intelligent Agent?

- An **intelligent agent** is an agent that selects actions in a way that is intended to achieve its objectives effectively OR
- An intelligent agent perceives its environment, processes the available information, and selects actions that help achieve its goals
- For example, a vacuum robot:
  1. Detects dirt
  2. Determine its location
  3. Decides where to move
  4. Moves towards the dirt
  5. Cleans it

Prepared By: Ehtisham Rasheed

The Basic Intelligent Agent Model



Agent Does Not Necessarily Mean Robot

- An agent does **not** have to be a physical robot
- Agents can be Physical Agent as well as Software Agent
- Physical agent
  - Robots
  - Autonomous vehicles
  - Drones
- Software agent
  - Recommendation systems
  - Chatbots
  - Email filtering systems
  - Trading systems

Prepared By: Ehtisham Rasheed

Software Agent Example

- Consider an email spam filter
- Environment
  - Incoming emails
- Sensor/input
  - Email content
- Agent processing
  - Analyze sender, words, links, patterns
- Action
  - Classify email as spam or not

Prepared By: Ehtisham Rasheed

Percept and Percept Sequence

- An agent doesn't necessarily receive the entire environment
- It receives information through its sensors called **percept**
- A **percept** is the information an agent receives from the environment through its sensors at a particular moment
- One percept alone may not be enough
- An agent may receive a sequence of percepts over time
- This is called **percept sequence**
- The complete history of what the agent has perceived can influence its decision
- This becomes important when we discuss more advanced agent architectures

Prepared By: Ehtisham Rasheed

Agent Function and Agent Program

- The agent function is the **ideal mapping** from percepts to actions
- The agent program is the **actual implementation** of that function
- In other words
- Agent function is what the agent should do
- The computer program that **implements what the agent should do**
- For example:

```
Agent Function:
If dirt detected → clean

Agent Program:
Python code implementing this rule
```

Prepared by: Ehtisham Rasheed

What is a Rational Agent?

- A **rational agent** chooses an action that maximizes its expected performance based on:
  1. The information it has perceived
  2. Its prior knowledge
  3. The actions available to it
  4. The expected consequences of those actions
  5. The performance measure used to evaluate it
- A simple definition:
  - A rational agent chooses the action expected to maximize its performance measure given the information available to it.

Prepared By: Ehtisham Rasheed

Intelligent Agent vs Rational Agent

- **Intelligent Agent** is the **broader concept**
- **Rational Agent** is an intelligent agent that specifically **chooses actions expected to produce the best performance** based on what it knows and perceives
- A rational agent can make a wrong decision because it may have:
  - Incomplete information
  - Noisy sensors
  - Limited knowledge
  - An uncertain environment

Prepared By: Ehtisham Rasheed

Intelligent Agent vs Rational Agent

- An intelligent agent is a system that perceives its environment and takes actions to achieve objectives.
- A rational agent is an intelligent agent that selects the action expected to maximize its performance measure, given its percepts, knowledge, and available actions.

- Intelligent Agent → Perceives + Acts
- Rational Agent → Perceives + Acts + Chooses the Best Expected Action

Prepared By: Ehtisham Rasheed

Rationality vs Perfection

- Perfect Agent
  - A hypothetical agent that always knows the correct answer and always makes the perfect decision
- Rational Agent
  - An agent that chooses the best action **based on the information available to it**
- Real AI systems are designed to be rational, not omniscient
- Omniscient
  - **Omniscient** means knowing everything
  - A rational agent does not need to know everything

Prepared By: Ehtisham Rasheed

What is a Performance Measure?

- How do we know whether an agent is performing well?
- A **performance measure** is a criterion used to evaluate how successfully an agent is achieving its objectives
- Possible performance measures for Vacuum Cleaner:
  - Amount of dirt removed
  - Time taken
  - Energy consumed
  - Number of unnecessary movements
- We might define objective as:
  - Keep the floor clean while minimizing time and energy consumption

Prepared By: Ehtisham Rasheed

Performance Measure Examples

Agent	Possible Performance Measure
Vacuum robot	Cleanliness, time, energy
Self-driving car	Safety, travel time, comfort
Chess AI	Winning the game
Spam filter	Correct classification
Recommendation system	Useful/relevant recommendations
Delivery robot	Successful and timely delivery
Medical AI	Diagnostic accuracy and safety
Chatbot	Helpfulness, correctness, user satisfaction

The performance measure depends on the purpose of the agent.

Performance Measure vs Goal

- These terms are related but not exactly identical
- **Goal:** What the agent wants to achieve
  - Example: Reach Islamabad
- **Performance measure:**
  - How we evaluate how well it achieved that goal
  - For example:
    - Did it reach the destination?
    - How long did it take?
    - How safe was the journey?
    - How much fuel was consumed?
- Goal describes the desired outcome; performance measure defines how success is evaluated

Prepared By: Ehtisham Rasheed

Example — Self-Driving Car

- **Agent:** Self-driving car
- **Environment:** Road and surrounding traffic
- **Sensors:** Camera, Radar, GPS, Other sensors
- **Percepts:** Vehicle ahead, Traffic light, Pedestrian, Road lane
- **Processing:** The AI system analyzes the information
- **Action:** Brake, Accelerate, Turn, Change lane
- **Performance Measure:** Safety, Travel efficiency, Passenger comfort, Energy efficiency

Prepared By: Ehtisham Rasheed

Another Example — Chess AI

- **Agent:** Chess AI
- **Environment:** Chess board and opponent
- **Sensors/Input:** Current board position
- **Percepts:** Current state of the game
- **Processing:** The AI system analyzes the information
- **Action:** Possible chess moves
- **Performance Measure:** Winning the game

Prepared By: Ehtisham Rasheed

Another Example — Chatbot

- **Agent:** Chatbot
- **Environment:** Conversation with the user
- **Sensors/Input:** User's message
- **Percepts:** Current message and conversation context
- **Processing:** The AI interprets the input
- **Action:** Generate a response
- **Performance Measure:** Relevance, Correctness, Helpfulness, User satisfaction

Prepared By: Ehtisham Rasheed

A Simple Intelligent Agent in Python

- Suppose we've a vacuum cleaner

```
def vacuum_agent(location, status):
    if status == "Dirty":
        return "Suck"
    elif location == "A":
        return "Move Right"
    else:
        return "Move Left"
```

- Now test it

```
print(vacuum_agent("A", "Dirty"))
```

- Output: Suck

```
print(vacuum_agent("A", "Clean"))
```

- Output: Move Right
- This is a very simple **rule-based agent**
- Is this really AI?